

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Circles

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01

ANSWER KEY · SAT QUESTION BANK

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Q1**B****B.** 48π

Choice B is correct. Since $\overline{PR}$ and $\overline{QS}$ are diameters of the circle shown, $\overline{OS}$, $\overline{OR}$, $\overline{OP}$, and $\overline{OQ}$ are radii of the circle and are therefore congruent. Since $\angle SOP$ and $\angle ROQ$ are vertical angles, they are congruent. Therefore, arc PS and arc QR are formed by congruent radii and have the same angle measure, so they are congruent arcs. Similarly, $\angle SOR$ and $\angle POQ$ are vertical angles, so they are congruent. Therefore, arc SR and arc PQ are formed by congruent radii and have the same angle measure, so they are congruent arcs. Let x represent the length of arc SR . Since arc SR and arc PQ are congruent arcs, the length of arc PQ can also be represented by x . It's given that the length of arc PS is twice the length of arc PQ . Therefore, the length of arc PS can be represented by the expression $2x$. Since arc PS and arc QR are congruent arcs, the length of arc QR can also be represented by $2x$. This gives the expression $x + x + 2x + 2x$. Since it's given that the circumference is 144π , the expression $x + x + 2x + 2x$ is equal to 144π . Thus $x + x + 2x + 2x = 144\pi$, or $6x = 144\pi$. Dividing both sides of this equation by 6 yields $x = 24\pi$. Therefore, the length of arc QR is $24\pi + 24\pi$, or 48π .

Choice A is incorrect. This is the length of arc PQ , not arc QR .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2**B****B.** $\frac{1}{3}\pi$

Choice B is correct. The measure of an angle, in radians, can be found by multiplying its measure, in degrees, by $\frac{\pi}{180}$. It's given that the measure of angle Z is 60° . It follows that the measure, in radians, of angle Z is $60\frac{\pi}{180}$, or $\frac{1}{3}\pi$.

Choice A is incorrect. This is the measure, in radians, of an angle whose measure is 30° , not 60° .

Choice C is incorrect. This is the measure, in radians, of an angle whose measure is 120° , not 60° .

Choice D is incorrect. This is the measure, in radians, of an angle whose measure is 180° , not 60° .

Q3**13**

The correct answer is 13. The equation of a circle in the xy -plane can be written in the form $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center of the circle and r is the radius of the circle. It's given that the circle in the xy -plane is defined by $(x + 2)^2 + (y + 5)^2 = 169$. Therefore, $r^2 = 169$. Taking the positive square root of both sides of this equation yields $r = 13$. Thus, the radius of the circle is 13.

Q4**81**

The correct answer is 81. The measure of an angle, in degrees, can be found by multiplying its measure, in radians, by $\frac{180 \text{ degrees}}{\pi \text{ radians}}$. Multiplying the given angle measure, $\frac{9\pi}{20}$ radians, by $\frac{180 \text{ degrees}}{\pi \text{ radians}}$ yields $\frac{9\pi}{20} \text{ radians} \cdot \frac{180 \text{ degrees}}{\pi \text{ radians}}$, which is equivalent to 81 degrees.

Q5**D****D.** 234

Choice D is correct. Since the endpoints of minor arc MN and major arc MPN are the same, and the arcs together form a full circle, the sum of the lengths of these two arcs is equal to the circumference of the circle. It's given that the length of minor arc MN is 39 centimeters and the length of major arc MPN is 195 centimeters. Therefore, the circumference of the circle, in centimeters, is $39 + 195$, or 234.

Choice A is incorrect. This is the length, in centimeters, of minor arc MN , not the circumference, in centimeters, of the circle.

Choice B is incorrect. This is the difference of the lengths of major arc MPN and minor arc MN , in centimeters.

Choice C is incorrect. This is the length, in centimeters, of major arc MPN , not the circumference, in centimeters, of the circle.

Q6**D****D.** $x + 4^2 + y - 5^2 = 25$

Choice D is correct. A circle in the xy -plane can be represented by an equation of the form $x - h^2 + y - k^2 = r^2$, where h, k is the center of the circle and r is the length of a radius of the circle. It's given that the circle has its center at $-4, 5$. Therefore, $h = -4$ and $k = 5$. Substituting -4 for h and 5 for k in the equation $x - h^2 + y - k^2 = r^2$ yields $x - (-4)^2 + y - 5^2 = r^2$, or $x + 4^2 + y - 5^2 = r^2$. It's also given that the point $-8, 8$ lies on the circle. Substituting -8 for x and 8 for y in the equation $x + 4^2 + y - 5^2 = r^2$ yields $-8 + 4^2 + 8 - 5^2 = r^2$, or $-4^2 + 3^2 = r^2$, which is equivalent to $16 + 9 = r^2$, or $25 = r^2$. Substituting 25 for r^2 in the equation $x + 4^2 + y - 5^2 = r^2$ yields $x + 4^2 + y - 5^2 = 25$. Thus, the equation $x + 4^2 + y - 5^2 = 25$ represents the circle.

Choice A is incorrect. The circle represented by this equation has its center at $4, -5$, not $-4, 5$, and the point $-8, 8$ doesn't lie on the circle.

Choice B is incorrect. The point $-8, 8$ doesn't lie on the circle represented by this equation.

Choice C is incorrect. The circle represented by this equation has its center at $4, -5$, not $-4, 5$, and the point $-8, 8$ doesn't lie on the circle.

Q7**B**

B. $(x+8)^2 + (y-6)^2 = 25$

Choice B is correct. An equation of a circle is $(x-h)^2 + (y-k)^2 = r^2$, where the center of the circle is (h,k) and the radius is r . It's given that the center of this circle is $(-8,6)$ and the radius is 5. Substituting these values into the equation gives $(x-(-8))^2 + (y-6)^2 = 5^2$, or $(x+8)^2 + (y-6)^2 = 25$.

Choice A is incorrect. This is an equation of a circle that has center $(8,-6)$. Choice C is incorrect. This is an equation of a circle that has center $(8,-6)$ and radius $\sqrt{5}$. Choice D is incorrect. This is an equation of a circle that has radius $\sqrt{5}$.

Q8**4**

The correct answer is 4. There are π radians in a 180° angle. An angle measure of 720° is 4 times greater than an angle measure of 180° . Therefore, the number of radians in a 720° angle is 4π .

Q9**C**

c. $\frac{2}{3}\pi$

Choice C is correct. Because points T, O, and P all lie on the x-axis, they form a line. Since the angles on a line add up to 180° , and it's given that angles POQ and ROT each measure 30° , it follows that the measure of angle QOR is $180^\circ - 30^\circ - 30^\circ = 120^\circ$. Since the arc of a complete circle is 360° or 2π radians, a proportion can be set up to convert the measure of angle QOR from degrees to radians: $\frac{360 \text{ degrees}}{2\pi \text{ radians}} = \frac{120 \text{ degrees}}{x \text{ radians}}$, where x is the radian measure of angle QOR. Multiplying each side of the proportion by $2\pi x$ gives $360x = 240\pi$. Solving for x gives $\frac{240}{360}\pi$, or $\frac{2}{3}\pi$.

Choice A is incorrect and may result from subtracting only angle POQ from 180° to get a value of 150° and then finding the radian measure equivalent to that value. Choice B is incorrect and may result from a calculation error. Choice D is incorrect and may result from calculating the sum of the angle measures, in radians, of angles POQ and ROT.

Q10**192**

The correct answer is 192. The measure of an angle, in degrees, can be found by multiplying its measure, in radians, by $\frac{180 \text{ degrees}}{\pi \text{ radians}}$. Multiplying the given angle measure, $\frac{16\pi}{15}$ radians, by $\frac{180 \text{ degrees}}{\pi \text{ radians}}$ yields $\frac{16\pi}{15} \text{ radians} \cdot \frac{180 \text{ degrees}}{\pi \text{ radians}}$, which simplifies to 192 degrees.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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